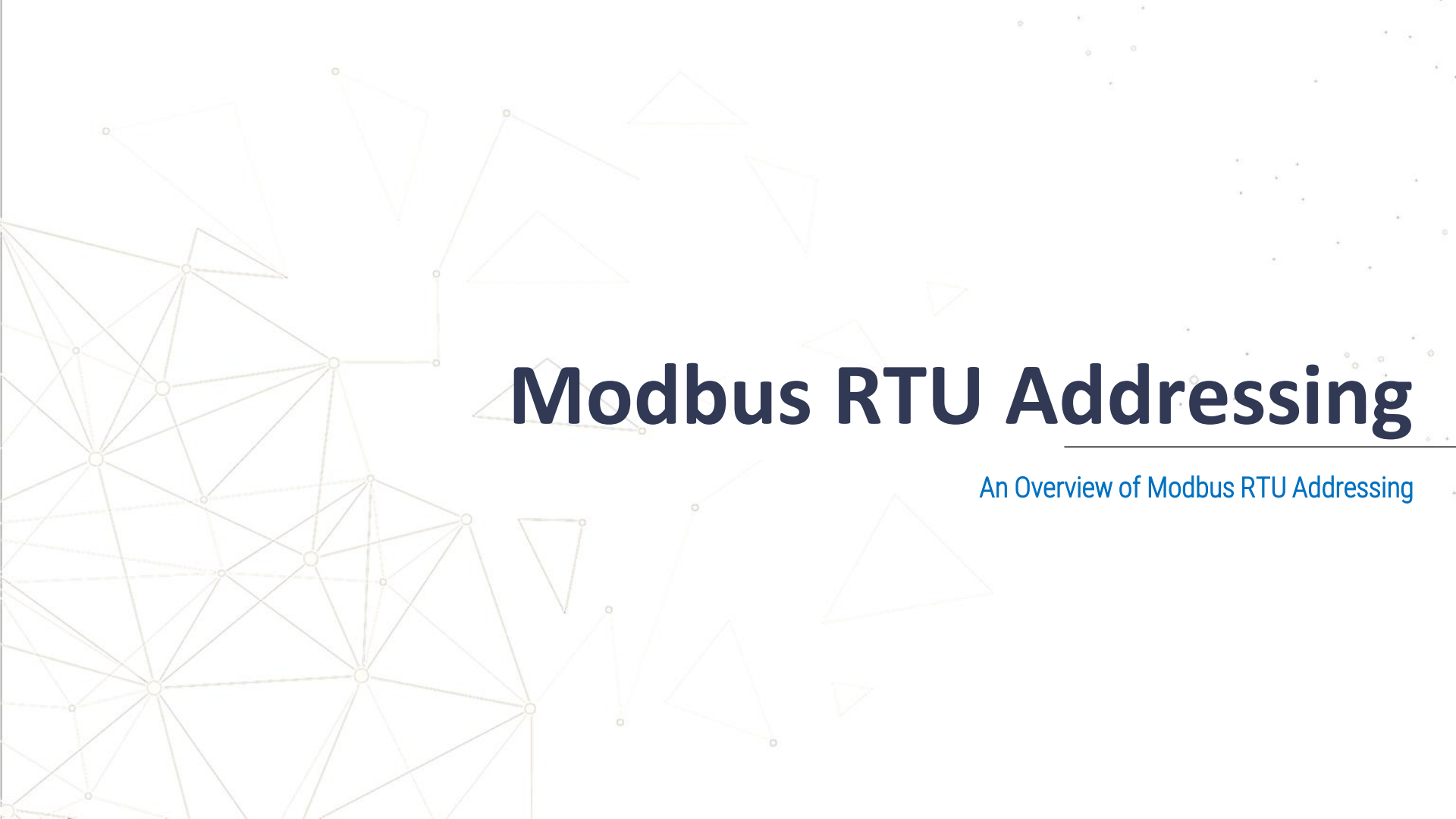




# Modbus RTU Addressing

An Overview of Modbus RTU Addressing

# Modbus RTU Addressing

- Modbus Register Types
- Modbus Register Types Address Ranges
- Logical vs. PDU Addresses
- Register Offsets Explained Further
- Addressing in Practice

# Modbus Register Types

## Four Types of Modbus Registers

- **Coils:** Binary outputs that can be controlled by the master device with two states: **ON (1)** and **OFF (0)**, **Read/Write**, used for controlling various processes.
- **Discrete Inputs:** Binary inputs provide status information to the master device. **Read Only**; often used for reading the state of sensors, switches, or device status.
- **Input Registers:** 16-bit registers that store analog data or data received from sensors. **Read Only**; updated with valuable information, such as, device readings.
- **Holding Registers:** 16-bit registers that are commonly used controlling slave devices. **Read/Write**, often used for configuration settings or process control variables.

# Modbus Register Types Address Ranges

## Memory Organization

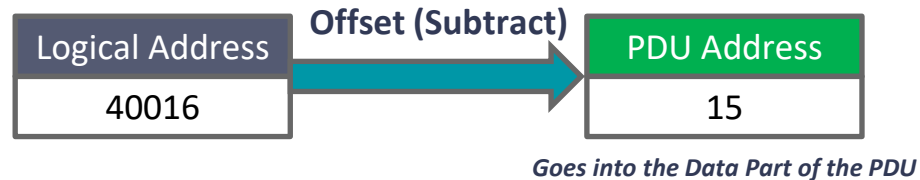
- **The Modbus Memory Area:** Divided into four types, each with specific address ranges. Coils and Discrete Inputs use 1-bit data, while Input Registers and Holding Registers use 16-bit data.

| Register Type     | Address Range | Data Type | Access          |
|-------------------|---------------|-----------|-----------------|
| Coils             | 00001 – 09999 | 1 Bit     | Read/Write (RW) |
| Discrete Inputs   | 10001 – 19999 | 1 Bit     | Read-Only (R)   |
| Input Registers   | 30001 – 39999 | 16 Bit    | Read-Only (R)   |
| Holding Registers | 40001 – 49999 | 16 Bit    | Read/Write (RW) |

# Logical vs. PDU Addresses

## Modbus Addresses Can be Represented in Two Ways

- **Logical Address:** Common in documentation, indicates the specific type of data using prefixes (e.g., 4XXXX for holding registers, 3XXXX for input registers).
- **PDU Address:** The actual address used during communication. Logical addresses include an offset, while PDU addresses start from 0.
- **Example:** 40016 (Holding Register where Offset = 40001) is sent as 15 in the PDU.



# Register Offsets Explained Further

**Offsets:** Are the difference between the logical representation of an address and its actual physical location in the PDU.

- **Example:** Documentation mentions **Holding Register 4014**; it's the logical address. When forming a Modbus request, this address will actually be sent as **13** (after subtracting the offset).

| Register Type     | Starting Address<br><small>(Can vary depending on manufacturer documentation)</small> |
|-------------------|---------------------------------------------------------------------------------------|
| Coils             | 0001                                                                                  |
| Discrete Inputs   | 1001                                                                                  |
| Input Registers   | 3001                                                                                  |
| Holding Registers | 4001                                                                                  |

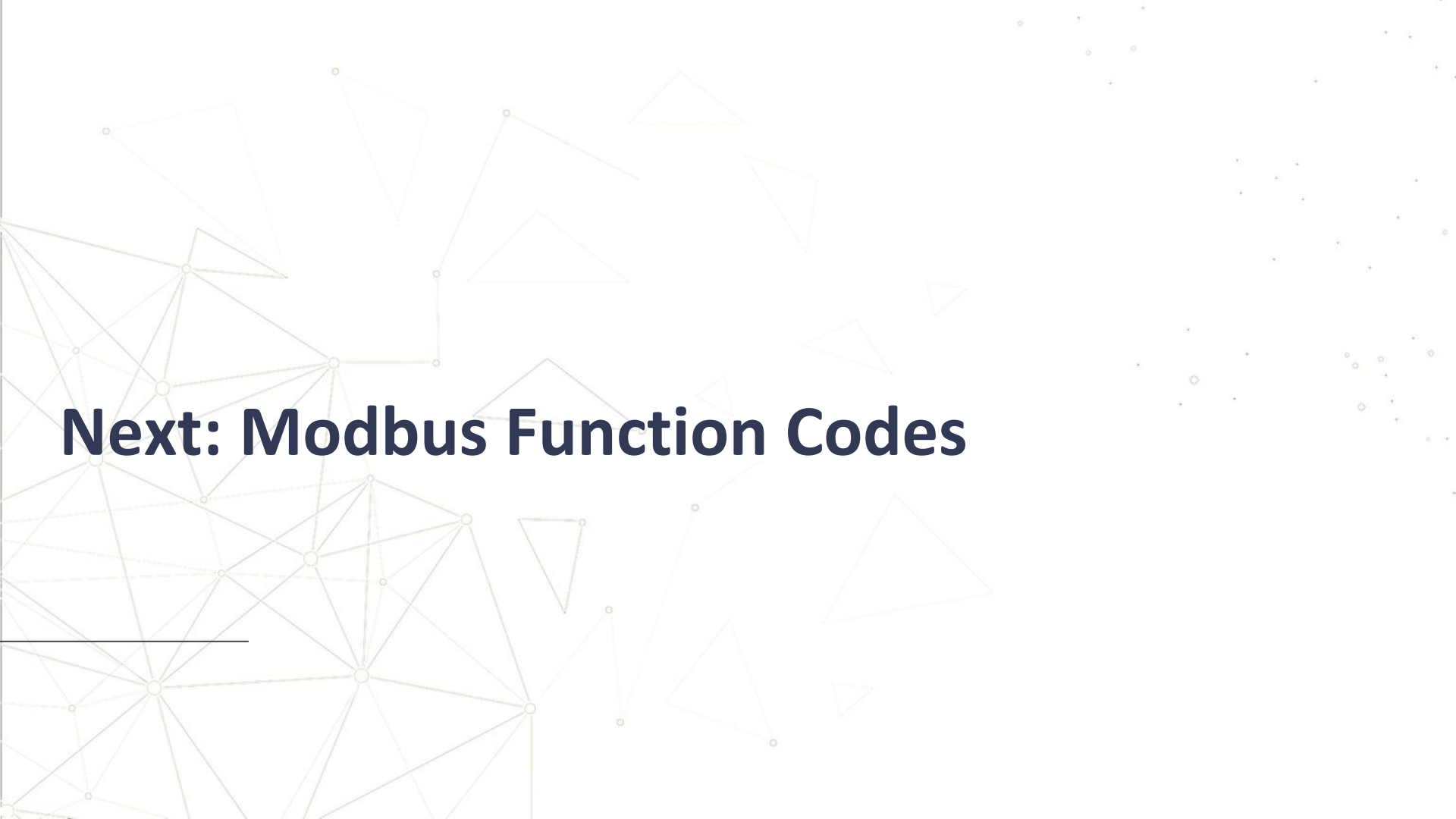
# Addressing in Practice

**Modbus Addresses RTU PDU in Further Detail:** The master sends a request specifying a **Slave Address**, **Function Code**, and the **PDU Register Address**.

- **Example:** The Modbus Master wants to read **Holding Register 4014** from a slave device with **address 7**, the Modbus RTU request will look something like this:
- 07 03 00 0D 00 01 [CRC]

| Slave Address | FC | Register Address | Number of Registers to Read | CRC     |
|---------------|----|------------------|-----------------------------|---------|
| 07            | 03 | 00 0D            | 00 01                       | 2 Bytes |

← Data Part of the PDU →

The background features a complex network of thin, light-colored lines and dots, forming various triangular shapes and a larger, interconnected web-like structure. The lines are primarily in shades of light green and yellow, with some darker green accents. The dots are small, light-colored circles. The overall effect is a subtle, technical, and geometric pattern.

# **Next: Modbus Function Codes**

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